

Validation of monoclonal antibody-based ligand binding assay for DDC as potential diagnostic biomarker for Parkinson's

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Abstract

The diagnosis of Parkinsonian disorders is currently based on clinical criteria. Recently, it was shown that increased levels of DOPA decarboxylase (DDC) in plasma and cerebrospinal fluid (CSF) can identify patients with Lewy body disease (LBD) and atypical Parkinsonian syndromes [1–4]. Although the effect of dopaminergic treatment (L-DOPA + DDC inhibitor) on DDC level remains elusive, DDC was also shown to be upregulated in de novo patients with LBD. These data were obtained using Olink technology, which relies on undisclosed poly- or monoclonal antibodies (Abs) and relative quantification.

To overcome these limitations, we have developed an in-house ligand binding assay (LBA) with Singulex/Erenna technology solely based on commercially available monoclonal Abs. Our primary aim is to validate the utility of our DDC LBA for diagnosis of treatment-naïve Parkinson's disease patients vs. healthy controls at baseline and compare results with visit 8, 3 years later (time and/or medication effect). Comparison to patients with scans without evidence of dopaminergic deficit (SWEDD) at baseline will provide additional insights into the specificity of DDC as a potential biomarker for dopaminergic deficit. This work will contribute to the potential establishment of DDC LBA as a promising diagnostic test with commercial Abs easily accessible to the research community.

Method

Capture- and detection antibodies were labelled using Capture/Detection Labeling Kits according to manufacturer's instructions (Merck; 03-0077-02 and 03-0076-02). Magnetic beads coupled with capture antibody were prepared in Coated Bead Buffer at 10 mg/mL and 17.5 µg IgG/mg beads. Detection antibody was adjusted to 1 mg/mL during labeling with fluorescent dye. Samples, calibrator (R&D Systems; 3564-DC) and antibodies were diluted in LowCross-Buffer MILD (Candor; 101 500). CSF samples were diluted 1:10. To run the Erenna assay, 100 µL of diluted sample or calibrator was added to a 96-well plate (Corning; P-96-450-C). Capture beads were diluted 1:200 and 100 µL was added to each well containing diluted sample or calibrator (5 µg beads/well). The plates were incubated for 2 h at 25 °C in a plate shaker (Boekel Scientific; 130000-115V). Beads were washed with Erenna System buffer (Merck, 02-0111-03) using an automated plate washer (Tecan; 30045769) and then resuspended with 20 µL/well of 937.5 ng/mL detection antibody. The plates were incubated for 1 h at 25 °C in the plate shaker.



After washing, the beads were resuspended and transferred into a new 96-well plate. The wash buffer was then removed from the beads in a final aspiration step. A total of 9.5 µL/well of Elution buffer B (Merck, 02-0297-00) was added to the beads for elution and incubated for 20 min at 25 °C in the plate shaker. The eluate was magnetically separated from the beads, transferred into a 384-well plate (Thermo Fisher Scientific; 264573) and neutralized immediately with 9.5 µL/well of Buffer D (Merck, 02-0368-00). After heat-sealing, the plates were analyzed using the Singulex Erenna instrument. Limits of quantification were estimated: LLOQ = 1 pg/mL; ULOQ = 316 pg/mL.

All CSF samples from PPMI subjects were subjected to one freeze-thaw cycle after receipt and before thawing for DDC analysis, since they were previously tested in PPMI project 173.

Four pooled CSF samples were included as quality controls on all four plates. Since the inter-plate coefficient of variation was below 6% for all quality control samples, no inter-plate normalization was performed. For one subject (SMS_SUBJECT_ID ST-00001272) no data is reported, since there was not enough volume left for a duplicate measurement.

References

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